

PRISM-4D GLP-1R / Aleniglipron Med-Chem Action Appendix

Medicinal-chemistry / SAR review of M2 replacements

Confidential Discussion Material

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Use of this document. This package presents computational translational evidence from the PRISM-4D engine and downstream analysis layers. It is *not* a clinical-effect claim, *not* a patient-response prediction, and *not* experimental validation. Assay recommendations are falsification gates, not confirmation ceremonies. PROJECTED and HYPOTHESIZED items require wet-lab falsification before any biological assertion.

Epistemic legend. Every claim, table row, and figure inherits exactly one of the following classes, which governs how it may be cited:

- **OBSERVED** direct tensor measurement from PRISM-4D engine outputs.
- **DERIVED** deterministic transform of observed tensors.
- **INFERRED** multi-tensor interpretation; not a single-tensor measurement.
- **PROJECTED** translational extrapolation beyond simulated conditions.
- **HYPOTHESIZED** requires wet-lab falsification before any biological claim.

1 Section 1 — Aleniglipron Interference Summary

1.1 Aleniglipron Thermodynamic Field Interference Analysis

1.1.1 Layer 1: Whole-Molecule Scoring

- Total Pi_clash across 9 critical edges: 41.8023
- Total Pi_complement: 46.1713
- Projected durability score: 546.6557 +/- 60.9397
- Confidence class: moderate

1.1.2 Layer 2: Fragment Decomposition (BRICS)

- Fragments identified: 8
- Dominant liability fragment: FRAG-C ([13*]c1c2c(nn1[19*])CCN([31*])[C@@H]2C, 20.2% of total clash)
- Inter-fragment coupling: 2.4970

1.1.3 Per-Edge Attribution

Edge	Whole Clash	Sum Fragment Clash	Coupling	Dominant Fragment	Dominant Fraction
glp1r_5VEX_WT:PHE143->ILE147	21.1308	19.7931	1.3376	FRAG-C	20.2%
glp1r_5VEX_WT:PHE143->TYR148	0.0000	0.0000	0.0000	FRAG-A	0.0%
glp1r_5VEX_WT:TYR148->TYR152	0.0000	0.0000	0.0000	FRAG-A	0.0%
glp1r_5VEX_WT:TYR152->TYR148	20.6716	19.5122	1.1593	FRAG-C	20.6%
glp1r_6LN2_A316T:TYR242->ILE313	0.0000	0.0000	0.0000	FRAG-A	0.0%
glp1r_6XOX_WT:ARG421->TRP417	0.0000	0.0000	0.0000	FRAG-A	0.0%
glp1r_6XOX_WT:PHE169->SER416	0.0000	0.0000	0.0000	FRAG-A	0.0%
glp1r_6XOX_WT:TRP417->ARG421	0.0000	0.0000	0.0000	FRAG-A	0.0%
glp1r_6XOX_WT:TYR242->ARG310	0.0000	0.0000	0.0000	FRAG-A	0.0%

1.1.4 Structural Rationales

glp1r_5VEX_WT:PHE143->ILE147: At glp1r_5VEX_WT:PHE143->ILE147, aleniglipron shows $Pi_clash = 21.13$. FRAG-C is the dominant contributor (4.26, 20% of total). Inter-fragment coupling contributes 1.34 additional clash, indicating synergistic steric compression when fragments are connected.

glp1r_5VEX_WT:PHE143->TYR148: Aleniglipron shows negligible steric interference at glp1r_5VEX_WT:PHE143->TYR148 ($Pi_clash = 0.000$). No fragment contributes significant clash.

glp1r_5VEX_WT:TYR148->TYR152: Aleniglipron shows negligible steric interference at glp1r_5VEX_WT:TYR148->TYR152 ($Pi_clash = 0.000$). No fragment contributes significant clash.

glp1r_5VEX_WT:TYR152->TYR148: At glp1r_5VEX_WT:TYR152->TYR148, aleniglipron shows $Pi_clash = 20.67$. FRAG-C is the dominant contributor (4.26, 21% of total). Inter-fragment coupling contributes 1.16 additional clash, indicating synergistic steric compression when fragments are connected.

glp1r_6LN2_A316T:TYR242->ILE313: Aleniglipron shows negligible steric interference at glp1r_6LN2_A316T:TYR242->ILE313 ($Pi_clash = 0.000$). No fragment contributes significant clash.

glp1r_6XOX_WT:ARG421->TRP417: Aleniglipron shows negligible steric interference at glp1r_6XOX_WT:ARG421->TRP417 ($Pi_clash = 0.000$). No fragment contributes significant clash.

glp1r_6XOX_WT:PHE169->SER416: Aleniglipron shows negligible steric interference at glp1r_6XOX_WT:PHE169->SER416 ($Pi_clash = 0.000$). No fragment contributes significant clash.

glp1r_6XOX_WT:TRP417->ARG421: Aleniglipron shows negligible steric interference at glp1r_6XOX_WT:TRP417->ARG421 ($Pi_clash = 0.000$). No fragment contributes significant clash.

glp1r_6XOX_WT:TYR241->ARG310: Aleniglipron shows negligible steric interference at glp1r_6XOX_WT:TYR241->ARG310 (Pi_clash = 0.000). No fragment contributes significant clash.

1.1.5 Recommendation

Prioritize modifications to FRAG-C if reducing receptor-side steric clash is the objective; protect fragments with low dominant clash unless potency SAR requires changes.

2 Section 2 — Zero-Shot Top-10 Replacements

SAR-contingency notice. Every replacement below is PROJECTED or HYPOTHESIZED. These are not synthesis instructions, not validated compounds, and not biological recommendations. Use as a SAR shortlist input only, subject to medicinal-chemistry review and wet-lab falsification.

3 ZeroShot_Top10_Replacements

Top-10 PROJECTED / HYPOTHESIZED fragment replacements derived from the manual emulation track. **These are not validated compounds and not synthesis instructions.** `solution_epistemic_class` governs how each row may be cited. SMILES strings are preserved in the CSV (omitted from MD for width).

solution_rank	anchor_id	sa_score	pi_complement	pi_clash	projected_durability_improvement	liability_edge_label	solution_epistemic_class
1	CAL-ANCHOR-000453	3.349	14.61	0	0.7327	PHE143 -> TYR148	PROJECTED
2	CAL-ANCHOR-000405	3.349	14.37	0	0.7269	PHE143 -> TYR148	PROJECTED
3	CAL-ANCHOR-000338	3.355	14.26	0	0.7241	PHE143 -> TYR148	PROJECTED
4	CAL-ANCHOR-000177	3.359	14.2	0	0.7225	PHE143 -> TYR148	PROJECTED
5	CAL-ANCHOR-000212	3.355	14.18	0	0.722	PHE143 -> TYR148	PROJECTED
6	CAL-ANCHOR-000142	3.359	14.17	0	0.7217	PHE143 -> TYR148	PROJECTED
7	CAL-ANCHOR-000107	3.359	14.16	0	0.7216	PHE143 -> TYR148	PROJECTED
8	CAL-ANCHOR-000473	3.349	14.12	0	0.7206	PHE143 -> TYR148	PROJECTED
9	CAL-ANCHOR-000302	3.089	14	0	0.7175	PHE143 -> TYR148	PROJECTED
10	CAL-ANCHOR-000373	3.355	13.99	0	0.7172	PHE143 -> TYR148	PROJECTED

4 Section 3 — Fragment Interference Attribution

5 Fragment_Interference_Attribution

Per-edge fragment interference attribution. Decomposes whole-molecule pi clash/complement into fragment contributions. INFERRED epistemic class — multi-tensor interpretation.

edge_id	whole_molec ule_clash	whole_mole cule_comple ment	sum_fragme nt_clash	sum_fragme nt_comple ment	dominant_fr agment	dominant_fr agment_clas h	dominant_fr action
glp1r_5VEX _WT:PHE14 3->ILE147	21.13	0	19.79	0	FRAG-C	4.261	0.2016
glp1r_5VEX _WT:PHE14 3->TYR148	0	25.5	0	23.83	FRAG-A	0	0
glp1r_5VEX _WT:TYR14 8->TYR152	0	20.67	0	19.51	FRAG-A	0	0
glp1r_5V EX_WT :TYR15 2->TYR148	20.67	0	19.51	0	FRAG-C	4.256	0.2059
glp1r_6L N2_A316 T:TYR24 2->ILE313	0	0	0	0	FRAG-A	0	0
glp1r_6XOX _WT:ARG4 21->TRP417	0	0	0	0	FRAG-A	0	0
glp1r_6XOX _WT:PHE16 9->SER416	0	0	0	0	FRAG-A	0	0
glp1r_6XOX _WT:TRP41 7->ARG421	0	0	0	0	FRAG-A	0	0
glp1r_6XOX _WT:TYR24 1->ARG310	0	0	0	0	FRAG-A	0	0

6 Section 4 — Phase 2D Staged Variant Grid

Materialization status. Phase 2D targets are *staged* — they appear in the planning manifest but have not been engine-executed. Treat the table below as a queue, not an evidence set.

7 Phase2D_Staged_Targets

Phase 2D variant-grid staged targets. **Materialization status:** `staged_manifest_materialized_not_executed` — these are *staged*, not engine-executed. The Phase 2D manifest is a planning artifact.

condition_id	variant	background	topology_source	evidence_status	epistemic_class	n_replicas	engine_run_id
glp1r_5VEX_WT		WT					
glp1r_6LN2_A316T		A316T					
glp1r_6LN2_T149M		T149M					
glp1r_6LN2_WT		WT					
glp1r_6X1A_WT		WT					
glp1r_6XOX_A316T		A316T					
glp1r_6XOX_T149M		T149M					
glp1r_6XOX_WT		WT					

Release identity.

Listing 1: Release identity (verbatim)

```
campaign_id      : glp1r_aleniglipron
release_version  : PRISM_GLP1R_M2_DELIVERABLES_v1.1
generated_at_utc : 2026-05-24T03:47:26Z
cbom_merkle_root : 1461e688bada66c7537bbce30e3b82ab1ec5f214e85cfb7d7ac966c60a7c1856
executive_archive_sha : b97bf1499fb82b6544c0a50f14e757ceef5e094d00c07f9f8c4633b87b219454
```